

# COEXTEND AI — PROSPECT INTELLIGENCE MVP

## AI Automation Engineer Candidate Interview Presentation & Blueprint

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Company: Coextend Global LLP (Pune, India | Target Markets: UK, US & Canada)

Assessment Charter: Paid Capability Assessment for AI Automation Engineer Role

**Core Stack: Python 3.14 | FastAPI | Vector RAG | Deterministic 100-Pt Scoring | Pydantic v2**

# Slide 2: Business Problem & Pre-Sales Founder Bottleneck

## WHY COEXTEND ASKED FOR THIS PROJECT

### 1. The Founder's Bottleneck

- Founder spends 30–45 mins per lead on manual research across LinkedIn, websites & feeds.
- At 100 leads/month, burns 50–75 hours of high-value founder time.
- Manual searches create qualification inconsistencies.
- Goal: Compress 45 mins into a 3-min decision-ready brief.

### 2. High-Stakes Façade Market

- Façade tenders range from £500k to £20M+ with strict 2-3 wk deadlines.
- Generic cold pitches fail and ruin credibility.
- Outreach must cite specific CWCT, BS EN 13830, ASTM & software tools.
- Goal: Arm sales team with exact technical pain points.

### 3. The Engineering Assessment

- Paid capability assessment for AI Automation Engineer role.
- Tests software architecture, RAG grounding, and deterministic math.
- Evaluates 5-day delivery roadmap execution.
- Goal: Deliver a clean, audited, and tested candidate build.

# Slide 3: Why Generic AI Prompts Fail & Architectural Solution

## THE ENGINEERING PROBLEM & ARCHITECTURAL SOLUTION

### 1. Hallucinated Capabilities

- Generic LLMs confuse civil drafting with specialist façade takeoffs.
- AI promises services Coextend doesn't offer (e.g., structural design).
- Uses US CSI codes for UK contractors who need Uniclass/NRM2.

### 3. Fabricated Facts

- LLMs invent contract wins and assume non-existent project scopes.
- Zero audit trail leaves sales reps vulnerable on calls.

### 2. Erratic 'Vibes' Scoring

- LLMs give 85/100 on Monday and 45/100 on Tuesday for the same data.
- Subjective scoring breaks CRM pipeline prioritization.
- Probabilistic math cannot be audited or trusted.

### 4. Two-Pillar Solution

- Pillar 1: Internal RAG grounded ONLY in approved Coextend IP.
- Pillar 2: Live Web Research with [Verified] / [Probable] tags.
- Scoring: 100% pure Python code—zero LLM math variance.

# Slide 4: Knowledge Base Integration (14 Shared Documents)

## MAPPING SHARED FILES TO SYSTEM LAYERS

### 1. RAG & System Prompt

- 01 - Agent Instructions: Evidence tagging ([Verified]/[Probable]) -> System Prompt.
- 01 - Company Profile: Grounded facts (Pune hub, 10 staff) -> Vector DB.
- 02 - Services & 03 - Capabilities: 8 service lines & CWCT standards -> RAG retrieval.

### 3. Scoring & Proposals

- 05 - ICP & Rubric: Revenue bands, geographic priority -> Pure Python scoring.
- 08 - Proposal Templates: 3 commercial engagement models -> Next Action.

### 2. Web Research & Pydantic Brief

- 02 - Research Criteria: 8 research categories & target roles -> Scraper blueprint.
- 04 - Research Output Template: 11-section layout -> Enforced Pydantic schema.
- 03 - Strategy & 04 - Positioning: Value props & objections -> Brief hypotheses.

### 4. Multi-Touch Outreach

- 02 & 03 - LinkedIn Connection & Pitch: Role-specific connection & pitch drafts.
- 04 & 05 - Email & Follow Up: 3-step cold email cadences & 4-touch follow-ups.

# Slide 5: Full Codebase Architecture & File Structure

## DIRECTORY BREAKDOWN & COMPONENT FUNCTION

### Core Backend (`prospect-intelligence/`)

- `main.py`: FastAPI REST endpoints, async task queue, worker loop.
- `models.py`: Pydantic schemas (Intake, ResearchBrief, ScoreBreakdown, CRM).
- `config.py`: Environment variables, API keys, threshold settings.

### Intelligence Engines

- `researcher.py`: Async web scraper, BeautifulSoup parser, URL tagging.
- `rag.py`: SentenceTransformers embeddings, vector search, citations.
- `scoring/engine.py` & `rubric.py`: Pure Python 100-point evaluator.

### Outreach, CRM & Storage

- `outreach.py`: Role-based email & LinkedIn generator.
- `crm.py` & `export_db_to_excel.py`: HubSpot JSON payload & Excel exporter.
- `database.py`: SQLite persistence for runs, briefs, and scores.

### UI & Test Suite (`ui/` & `tests/`)

- `ui/`: FastAPI Jinja2 templates, `manifest.json`, `service-worker.js` (PWA).
- `static-demo/`: Standalone offline interactive HTML dashboard.
- `tests/`: Unit & integration test suite (anti-fabrication, scoring, CRM adapter).

# Slide 6: Deterministic 100-Point Lead Scoring Engine

## OBJECTIVE QUALIFICATION & ICP FACTORS

**10 Weighted ICP Factors (100 Pts Total)**

- 1. Trade Fit (20 pts): Façade specialist = 20, Roofing = 16, Main contractor = 10.
- 2. Tender Volume (15 pts): >=3 tenders = 15, 1-2 projects = 10, None = 0.
- 3. Geography (10 pts): UK = 10, US/Canada = 8, Europe/ANZ = 4, Other = 0.
- 4. Company Size (10 pts): Tier 1 (£20M+) = 10, Tier 2 = 8, Tier 3 = 5.
- 5. Estimating Need (10 pts): Tight deadlines/overflow = 10, Regular = 7.
- 6. Drafting/BIM Need (10 pts): BIM L2 / shop drawing demand = 10, CAD = 7.
- 7. Hiring Trigger (10 pts): Recruiting Estimators/QS = 10, Expansion = 4.
- 8. Decision-Maker Access (5 pts): Pre-Con Director = 5, MD/Owner = 4.
- 9. Outsourcing Readiness (5 pts): Uses technical subconsultants = 5.
- 10. Commercial Fit (5 pts): Prompt payment = 5, Standard = 3, High risk = 0.

**Priority Bands & Pure Code**

- HIGH PRIORITY (>= 70 Pts): Founder outreach & pilot takeoff pitch.
- MEDIUM PRIORITY (45–69 Pts): Standard email & LinkedIn connection.
- LOW PRIORITY (< 45 Pts): Nurture sequence or disqualify.
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- Pure Python: Zero math hallucination; 100% audit trail and repeatability.

# Slide 7: Grounded RAG & Anti-Fabrication Engine

## TWO-PILLAR RETRIEVAL & EVIDENCE STANDARDS

### Internal RAG Architecture

- Ingests 14 Coextend knowledge base PDFs into vector embeddings.
- Heading-aware chunking maintains section & document context.
- Top-k semantic retrieval injects exact capability snippets into LLM prompts with source metadata.

### External Research & Tagging

- Crawls company site, leadership, and public tender announcements.
- Tags findings with confidence tiers: [Verified] (direct URL), [Probable] (indirect), [Unverified].
- Every claim in brief includes clickable source citation.

### Strict Anti-Fabrication Constraints

- System prompt explicitly forbids guessing missing data.
- If data point cannot be verified, brief outputs 'No evidence found'.
- Eliminates false claims on sales calls and protects commercial reputation.

# Slide 8: Technology Stack & Production Engineering

## FASTAPI, PYDANTIC V2 & RESILIENCE

### Production Stack Breakdown

- Backend: Python 3.14 + FastAPI (Async REST API).
- Validation: Pydantic v2 (Strict typing, regex URL checks).
- Vector Engine: ChromaDB / FAISS + SentenceTransformers.
- Document Engine: ReportLab 5.0 (PDF) + python-pptx.
- Persistence: SQLite + JSON columns + Excel exporter.

### Resilience & Rate Limit Handling

- Async Scraper: Handles HTTP timeouts with exponential backoff.
- Cooldown: Enforces 24-hr domain scraping cooldown.
- Graceful Fallback: Scraper failure falls back to RAG mode.
- Structured Logging: All pipeline steps log detailed tracebacks.



# Slide 9: Testing Suite, QA & Safety Guardrails

## SYSTEM VERIFICATION & COMPLIANCE RULES

### Comprehensive Test Suite Coverage

- `test_anti_fabrication.py`: Validates zero-guess enforcement.
- `test_scoring.py`: Verifies scoring across High/Med/Low leads.
- `test_crm_adapter.py`: Ensures HubSpot JSON payload validity.
- `test_failure_injection.py`: Tests network timeout resilience.
- `test_e2e_prospects.py`: End-to-end integration test suite.

### Safety & Compliance Guardrails

- 1. No Automated LinkedIn Actions: Scraping & auto-sending strictly banned.
- 2. Human-in-the-Loop: Outreach remains 'draft' for review.
- 3. CRM Sandbox: Staged locally to protect live CRM data.
- 4. Strict Grounding: Service claims restricted to ingested files.

# Slide 10: Key Technical Interview Questions & Answers (1/2)

## DEFENDING ARCHITECTURAL DECISIONS

### Q1: Why separate RAG from Web Research?

- RAG holds Coextend internal IP; Web research holds external prospect data.
- Mixing them leads to LLM confusing Coextend services with prospect capabilities.
- Guarantees 100% accurate service matching.

### Q3: How do you prevent AI Hallucinations?

- System prompt anti-guess policy ('No evidence found').
- Grounded RAG with exact document citations.
- Pydantic schema validation for all LLM outputs.

### Q2: Why Python for Lead Scoring, not LLM?

- LLMs yield erratic, probabilistic 'vibes' scores.
- Python code guarantees 100% mathematical consistency, auditability, and zero API cost.

### Q4: How does CRM Export work?

- `crm.py` maps research brief to HubSpot Company & Contact schema.
- Includes domain deduplication and exports JSON/CSV staged payloads.

# Slide 11: Key Technical Interview Questions & Answers (2/2)

## DEFENDING ENGINEERING TRADE-OFFS

### Q5: How do you handle anti-scraping / rate limits?

- Browser headers, rate-limiting delays, timeout bounds, and 24-hr domain cooldowns.
- Scraper failure falls back gracefully to basic domain analysis without crashing.

### Q7: Why strictly avoid LinkedIn automation?

- LinkedIn aggressively detects automated actions, risking domain & account bans.
- Keeping outreach human-in-the-loop ensures 100% safety and high conversion quality.

### Q6: Why FastAPI and Pydantic v2?

- FastAPI provides native async support for non-blocking I/O.
- Pydantic v2 provides C-compiled fast validation and strict schema enforcement.

### Q8: What would you build in Phase 2?

- Real-time webhook sync with HubSpot/Salesforce.
- Automated tender portal monitoring (Tenders Direct, Contracts Finder).
- Multi-agent parallel web research for enterprise accounts.

# Slide 12: 5-Day Delivery Roadmap & Final Business Impact

## MILESTONE EXECUTION & ROI

**5-Day Milestone Delivery Schedule**

- Day 1: Architecture blueprint, repository skeleton, Pydantic schemas, FastAPI /health.
- Day 2: PDF RAG vector ingestion + web research engine tested on initial prospects.
- Day 3: Deterministic lead scoring engine + brief generator + CRM export adapter.
- Day 4: Outreach generator + UI dashboard + resilience logging & error handling.
- Day 5: 5-prospect evaluation suite, unit tests, master documentation & candidate presentation.

**Business Impact & Value Delivered**

- Time Saved: 90%+ reduction in pre-sales research time (45 mins -> 3 mins per lead).
- Scale: Enables founder to evaluate 500+ prospects/month easily.
- Accuracy: 100% grounded technical citations with 0% capability hallucination.
- Pipeline Quality: Objective 100-point scoring focuses sales team on High-Priority leads.
- Status: System architecture, verification docs & candidate presentation 100% complete.